# Boulder County Health Department Air Inspection Report

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SEP 15 1999
Air Pollution Control Division
Stationary Sources Program

**COUNTY NUMBER:** 013 **SOURCE NUMBER:** 003 **DATE:** 06/23/99

**COMPANY:** Southdown Inc. *(formerly Southwestern Portland Cement)*

**INSPECTOR:** Gabi Hoefler, Joni Rix

**SITE LOCATION:** 5134 Ute Hwy.., Lyons
**CONTACT PERSON:** Steve Mossberg, John Lohr
**TELEPHONE NUMBER:** 303-823-2113

**COUNTY:** Boulder
**TIME:** 9:00 am
**PERMIT NUMBER:** 12BO444-1&2, 93BO1414F, 98BO593, 98BO0259,98BO0315, & 98BO0292

**INSPECTION TYPE: Routine [ x ] Major Complaint [ ]** **FA [ ]** **Surveillance [ ]**
**TOTAL HOURS REQUIRED:** 49.0
**Prep/Travel:** 28.0
**Inspection:** 7.0
**Report:** 14.0
**Compliance Assist: [n] y/n** **P2 Assistance: [n]y/n**

**Emission Points Subject to NSPS/MACT?:** Yes - NSPS Subpart F & NSPS Subpart OOO

This was an announced major source inspection as per CDPHE-APCD/BCHD 1999 Air Contract. The inspection was conducted by myself, Joni Rix of BCHD, and Cindy Reynolds of the EPA. A tour of the plant was lead by Steve Mossberg, Facility Compliance Manager for Southdown and John Lohr, Plant Manager. The source is a dry process, Portland cement manufacturing facility. The plant was not in full operation at the time of inspection and a follow-up visit was required (see below). The CEM is certified and fully operational. Quarterly EER reports are submitted to CDPHE and all upsets are reported to BCHD and to CDPHE.

The Dowe Flats Quarry is now in full operation. Mining, hauling and crushing takes place at the site and the raw materials are transported via an enclosed conveyor belt to the plant site where it is dropped onto stockpiles from the material stacker. At the time of inspection, the stacker was inoperable due to high winds the day before. On 6-22-99, high winds reaching up to 60 mph tore the support mast from the frame of the stacker causing the entire structure to topple. The dust control equipment was not damaged so the plant was in the process of locating a replacement stacker to keep the plant running. Without a stacker, material cannot be crushed at Dowe Flats and conveyed to the Lyons site, hence none of this equipment was operating at the time of inspection.

A follow-up visit was made on 7-19-99 to observe the site in full operation and take any necessary visible emission readings. At the time of the inspection, the dryer was running 130 tons of limestone and the Dowe Flats conveyor belt and stacker were running 340 tons of limestone. The rental stacker was equipped with the old baghouse and was working well. There were no visible emissions from the stacker in full operation, only the mist from the water spray bars was visible. There were also no emissions from the primary crusher at the Lyons Quarry and only 5% emissions for approximately 10 seconds was noted from the front end loader dumping material into the crusher. The Dowe Flats Quarry was also in full operation at the time of the follow-up visit. There were no emissions observed from the primary crusher and emissions were contained within the enclosure or crusher housing when loaded by the large dump trucks. The haul road into the pit where mining was taking place was fairly dusty. There were no off-property emissions. John Lohr immediately called the water truck operator to come and water the road. The installation of the new sprinkler at Dowe Flats will take care of this problem. At the time of the visit, a crew was installing the sprinkler line along the ridge of the quarry and haul roads. This will give Southdown the flexibility to water the pit being mined or haul roads when necessary.

The only area of concern during the follow-up visit was the A-frame building where clinker is stored. Approximately 5% emissions were noted coming from the truck entrance/doorway to the A-Frame building. The dust was apparently coming from the clinker piles when material was dropped from the conveyor belt above. While the door must remain open for trucks to enter, Southdown was instructed to correct the problem immediately. Attached are the operating procedures Southdown has changed in order to correct this problem. They will now attempt to drop the clinker away from the door and into a pit so dust will not come off the piles. Additionally, Steve Mossberg has put in a request to replace the curtains/ flap doors on the entrance to the building. New management practices specify that if windy conditions persist, the clinker shall be put in the pit and then hauled to the A-Frame using the loader. See Attached " Plant Operation "A" Frame Clinker Storage, Use and Operation." A follow-up visit was made 08-11-99 and no visible emissions were noted from the A-frame or from the plant.

Action #025

There have been a number of changes since the last inspection in 1998. Due to a number of citizen complaints and housekeeping problems, Southdown was required by APCD, specifically Dave Ouimette, to revise their Fugitive Dust Control Plan (FDCP) in March and has implemented a number of changes in an attempt to control fugitive emissions. First, reclamation of the Lyons Quarry is well under way to offset the addition of the high efficiency separator. Native grass is now sprouting in the previously exposed area. They have also begun adding a citrus based foamer from the transfer points onto the stacker to better suppress fugitive emissions. As of this Spring, Southdown is now adding a surfactant addition to their water truck which is expected to extend the effective time of watering. Southdown has agreed to apply magnesium chloride at least three times this summer to the haul roads and has one vacuum sweeper and three sweepers on site 8 hours a day to clean-up the dirt and dried mud on the roadways.

Also in response to a number of citizen complaints, Southdown has revamped the truck wash area. The concrete driveway has been widened in the approach to the wash hoop and concrete barriers have been set up to channel all trucks through the wash hoop rather than allowing a choice of path. The truck wash has been set up to automatically turn on when a truck passes through the hoop and all trucks will be channeled through the wash as they exit the plant. Additionally, Southdown has reduced the use of 3rd ridge shale using more high calcium limestone. While they have to dig deeper to obtain the limestone, this process greatly reduces the generation of cement kiln dust (CKD)- a problem area in the past. Most days they produce no CKD and have reduced production to approximately 30,000 tons of CKD a year. Their goal is to produce less than 15,000 tons/year of CKD. In addition to the sprinkler system installed at the CKD disposal pit, Southdown is installing a sprinkler system at Dowe Flats to reduce fugitive emissions from quarry operations.

Southdown has begun to amend their record keeping system to prepare for their Title V Operating Permit which is expected to be finalized in September/October of this year. They are also developing a database to produce year to date information and compliance reports as well as track work orders and daily inspections. They continue to work on ensuring their systems are Y2K compliant.

Southdown has installed a high efficiency separator and a products collection baghouse in the finish cement mill system replacing two existing separators. The high efficiency separator will help achieve a better quality of cement, and at the same time, avoids over grinding of the cement. The company has also revised their permit to increase the production/handling of raw materials from the Dowe Flats Quarry. While these changes constitute a major modification of a major source, the company has proposed several other changes which result in an emissions decrease. These changes include: an underground reclamation system for handling raw material from stockpiles, reclamation of disturbed areas to minimize the emissions due to wind erosion, reduction of haul distance for disposal of cement kiln dust and raw materials waste dust by discontinuing hauling to Pit "D" and disposal only occurring at the closer C-Pit. In connection with these changes, three new permits have been issued and two existing permits were modified. Many of the emission sources which were previously grandfathered, are now covered under one of these new permits to render the change federally enforceable, and APCD will consider the overall modification as a synthetic minor modification of a major stationary source.

**Summary Status of New/Modified Permits:**
**Permit #93-BO-1414-F (Modified)** for the operation of the Dowe Flats Quarry was Self-Certified by Steve Mossberg in January 1999.

**Permit #94-BO-593 (Modified)** for the Belt Conveyor System and Crusher at Dowe Flats was Self-Certified by Steve Mossberg in January 1999 and a performance test for particulate matter was conducted in December 1997.

**Permit #98-BO-0259** a system wide permit for the Finish Mill began operation on March 23, 1999. Performance and opacity tests are scheduled for July 25, 1999. The permit must be self-certified by Sept 23, 1999 or 180 days from start-up date.
This permit cancels Permit #84BO369F for outdoor clinker storage.

**Permit #98-BO-0292** for Discharge from the Stacker, Underground Reclamation, and Handling by Front-End Loaders is pending construction. A request for the underground reclamation system was made on April 27, 1999 and approval from Southdown headquarters is still pending. This permit will be self-certified 180 days from start-up.

**Permit#98-BO-0315** for Storage, Handling, Processing, Haulage and Disposal of Cement Kiln Dust and Raw Material Waste Dust has not began operation. The new proposed "Pug Mill" for this operation is pending construction. This permit will be self-certified 180 days from start-up.

<u>The points of emission at the facility are as follows:</u>

| Point # | Point Description | Permit |
|---|---|---|
| *01* | *Primary Crusher* | *grandfathered* |
| *02* | *Raw Material Dryer* | *12B0444-1* |
| *03* | *Secondary Crusher* | *grandfathered* |
| *04* | *Raw Material Storage Silo* | *grandfathered* |

| | | |
|---|---|---|
| *05* | *Raw Mill (grinding)* | *grandfathered* |
| *06* | *Homogenizing / Blending* | *grandfathered* |
| *07* | *Precalciner / Kiln Process* | *12B0444-2* |
| *08* | *Clinker Cooler* | *12B0444-2* |
| *09* | *Clinker Storage Silos* | *98BO0259* |
| *10* | *Clinker Conveying / A-Frame Storage* | *98BO0259* |
| *11* | *Finish Mill & Bucket Elevator* | *98BO0259* |
| *13* | *Cement Storage Silos/Packout/Loadout* | *98BO025914* |
| | *Coal Unloading* | *exempt* |
| *15* | *Clinker Storage - outside* | *98BO0259* |
| *23* | *Wind Erosion* | *exempt* |
| *24A* | *Discharge from Stacker Conveyor Belt & Underground Reclamation (under construction)* | *98BO0292* |
| *24B* | *Handling of materials by front-end loader (Materials not handled by underground....)* | *98BO0292* |
| *25* | *Operation of Dowe Flats Quarry* | *93BO1414F* |
| *26* | *Belt Conveyor System for Dowe Flats* | *94BO593 -1* |
| *27* | *Primary Crusher Dowe Flats* | *94BO593 - 2* |
| *31* | *High Efficiency Separator & Baghouse Collector* | *98BO0259* |
| *39* | *Finish Mill System (consists of 6 emission points)* | *98BO0259* |
| *40A* | *Pneumatic Conveyance of CKD* | *98BO0315* |
| *40B* | *Pneumatic Conveyance of Benefication Dust* | *98BO0315* |
| *41* | *Pug Mill Mixing, Pelletization & Truck Loading of CKD* | *98BO0315* |
| *42* | *Haulage & Disposal of CKD & Benefication Dust* | *98BO0315* |
| *49* | *Storage, Handling, Hauling, Disposal of CKD (consists of 3 emission points)* | *98BO0315* |

*It should be noted that even though some points are grandfathered or exempt, they are still subject to all applicable standards and regulations of the Air Quality Commission including: visible emissions shall not exceed 20% opacity and no off-property transport. Many of the previously grandfathered sources are now covered under one of the revised or new initial permits.*

## FACILITY INSPECTION & PROCESS DESCRIPTION

**Point 01, Primary Crusher-Lyons Plant**

The primary crusher is the first piece of equipment in the process. It is on the south end of the plant, closest to the pits and is loaded directly through a hopper by front end loaders. The primary crusher is not used for material conveyed from the Dowe Flats quarry since the material is already crushed by the primary crusher at Dowe Flats but it is still used for the material remaining from the now completed Lyons quarry. Mining has now ceased at the Lyons Quarry, but some material previously mined remains to be crushed. The primary crusher was in operation during the inspection. Dust control is by a micro pulse baghouse. The baghouse is automatically cleaned by pulses of air that are triggered when the magnahelic pressure (pressure differential across the various bags) reaches a threshold number. A conveyor belt system then conveys the crushed rock (4" in diameter or less) to the dryer.

**Point 02, Raw Material/Limestone Dryer:** ***(Permit #12B0444-1)***

The dryer is essentially a small kiln or drum heater and is used to dry the material before it is sent on to the secondary crusher. The raw material is generally moist and even more so in the winter. There were no visible emissions at the time of the inspection. After the material is dried to remove most of the moisture, it is taken back to the secondary crusher.

**Compliance with Permit #12BO444-1 Conditions:**

1. Annual operating hours do not exceed 7,000 hours/year. Actual hours in 1998 were 6,669.5 hours.

2. Natural gas is the primary source of fuel and there was no coal burned in the dryer in 1998.

3. Emissions:

**Table 1.0**

| Pollutant | Average lbs/hr | Max. lbs/hr* | Permit Tons/yr | Actual Tons/yr* |
|---|---|---|---|---|
| $NO_x$ | 4.00 | 7.13 | 13.9 | 13.02 |
| $SO_2$ | 9.11 | 19.42 | 36.7 | 30.39 |

| | | | | |
|---|---|---|---|---|
| **VOC** | 41.66 | 72.38 | 144.8 | 138.92 |
| **CO** | 16.27 | 18.02 | 57.3 | 54.26 |
| **TSP** | 6.51 | 6.5 | 22.8 | 21.71 |
| **$PM_{10}$** | 6.51 | 6.5 | 22.8 | 21.71 |
| **Pb** | 0.13 | 4.81** | 15.8 | 0.43** |

* As per a Memo from Tom Tstinic from APCD, the permit engineer *(see attached)*, the lbs/hr limits for TSP, $SO_2$, $NO_x$, CO and VOCs were based on a stack test. Average hourly results were then adjusted by adding a 20% margin of error. Annual limits in tons/yr were then calculated based on 7000 hours per year. Maximum lbs/hour are the highest values of the three runs. Lead emissions are based on AP-42 emission factors. He says that *"compliance should be determined by using hours of operation, dryer feed (thruput) and natural gas consumption"*. The permit also states that *"the annual emission limits for this source have been calculated on the basis of the hourly average, hourly maximum and annual feed and fuel consumption rates shown below and 7,000 hours of operation per year." (See Table 2.0, below, for hours, feed and consumption)*. Therefore, compliance should be determined by comparing actual hours of operation, tons of dryer feed and natural gas consumption *(see Table 2.0, below)* not by comparing actual emissions in tons/yr or lbs/hr.

**The 4.81 lbs/hr of Lead in the permit was based on old AP-42 emission factors and was greatly over-estimated at the time the permit was written. In reality, the limit for lbs/hr should have been 0.13 lbs/hr. *(As per Jeet Gill of Southdown.)* Based on new EF's in AP-42 (75 x $10^{-5}$ lbs/ton of clinker) and research done by Southdown at their Knoxville facility that showed that 99.958% of the total Lead entering the kiln is retained in the kiln, the emission factor for Lead would be 0.00013 lbs/ton of dry feed. To be conservative, Southdown has used 0.001 lbs/ton of feed as an EF for Lead.

**Table 2.0**

| Description | Units | Hr. Average | Hr. Max | Annual Max | Actual Annual |
|---|---|---|---|---|---|
| ***Hours of Operation*** | Hours | n/a | n/a | 7,000 | 6,669.5 |
| ***Dryer Feed*** | Tons | 150 | 160 | 1,050,000 | 868,234 tons/yr or 130.2 tons/hour |
| ***Heat Input*** | MMBTU | 30 | 35 | 0.21 x $10^6$ | 0.18 x $10^6$ |
| ***Natural Gas*** | SCF | 29,903 | 41,860 | 209.32 x $10^6$ | 188.93 x $10^6$ |
| ***Backup Fuel:Coal*** | Tons | 1.3 | 1.4 | n/a | 0 |

4. The facility and control equipment are maintained and operated in a manner consistent with good air pollution control practices to minimize emissions. This was determined based on monitoring data and inspection of the facility.

5. Records of daily dryer feed rate, fuel feed rates and operating hours are maintained. Monthly summaries are prepared and kept on file for two years (or more) and include raw material consumption and quantities of natural gas (or coal) burned.

6. The permit number is marked on the equipment.

7. Opacity of gases did not exceed 10%. There were no visible emissions at the time of inspection. There was no concealment of emissions and records of startups, shutdowns, and malfunctions are maintained.

8. There have been no relaxation of any permit conditions, therefore, the source has not become major and PSD does not apply at this time.

9. SWP has calculated a rolling 12-month total of emissions on the first day of each month *(please see attached "Limestone Dryer Rolling Monthly Emissions")*. They keep a compliance record on site and also calculate monthly emissions. The emission factors used were calculated from allowable emissions (based on the stack test) and a feed rate of 130.2 tons/hr. The spreadsheet labeled "Limestone Dryer" *(attached)* shows these emission factors subtitled as "Calculated Emission Factors".

10. A revised APEN will be filed for any significant change in emissions.

11. Source has not become major since no permit conditions have been relaxed and are not subject to Reg. 3, IV.D.2.

12. SWP has notified the Division and BCHD of any upset conditions within 2 hours of the next working day with an explanation as to the cause and of the actions made to correct and prevent the problem.

**Compliance Status: [ X ] In compliance [ ] Out of Compliance**

**Point 03, Secondary Crusher**

There were no visible emissions noted during the inspection. The material being crushed is transported to the crusher via an enclosed belt system. This crusher crushes the rock to ½" diameter. From here the material is transferred to the raw material silos. Waste dust is collected by a baghouse and is then loaded through a drop chute into trucks. The waste dust loadout area was a significant problem in the past. SWP has since added waterline pressure to the pug mill which allows for better pugging. This additional water pressure greatly improves the wetting of the dust. They also re-configured the loadout area by moving the loadout to the outside, constructing a loadout platform and transferring material via a drop chute so there is much less fugitive emissions during truck loading. Southdown is working on installing a twin screw pug mill in 1999 which mixes better and will make the clinker dust pellets more compact, reducing fugitives. Southdown has reduced the use of third ridge shale, using more high calcium limestone, generating less CKD. They produce CKD only a couple days a month now, totaling 30,000 tons per year. They have reduced their production of CKD from 150,000 tons per year and intend to produce less than 15,000 tons per year using the limestone.

**Point 04, Raw Materials Silos**

The crushed and dried materials are then conveyed to and stored in silos which also store silica, iron ore and gypsum. In 1995, three new dust collectors were installed and the screw discharge was refined. There was no fugitive dust seen from this point during inspection.

**Point 05, Raw Materials Grinding (Raw Mill)**

This is a ball-mill system which uses steel balls to grind the raw material. The mill is actually a horizontal cylinder, 24 feet long and 14 feet in diameter, that is filled with thousands of steel balls. There are ribs or blades that run the entire length of the mill which carry the balls up to the top of the cylinder. As the mill turns, the balls tumble on the material and crush it into a fine powder of which 70% passes through a standard 200 mesh sieve. Pneumatic blowers transfer the ground material to the homogenizing silos. Air separators separate fine materials from coarse materials and re-circulate the coarse materials into the mill. There were no emissions observed at this point during the inspection. The particulate emissions are controlled by a baghouse.

**Point 06, Homogenizing/Blending Silos**

The homogenizing silos blend the ground raw material with and then the aerated material is fed into a separate silo prior to entering the precalcining process. There were no visible emissions observed from this point. Dust collectors collect particulates prior to entering and after exiting the homogenizing silos.

**Point 07 & 08, Precalciner-Kiln & Clinker Cooler** ***(Permit #12B0444-2)***

The preheater or precalciner tower supports a series of vertical cyclones through which the raw kiln feed passes on its way to the kiln. The hot gases rising from the kiln heat the material as it swirls through the cyclones and remove alkali, sulfur and hydrocarbons. The precalciner is on the tower above the feed end of the kiln . This is where the calcination process occurs or the removal of Carbon Dioxide ($CO_2$) from Calcium Carbonate ($CaCO_3$) to form Calcium Oxide (CaO). As much as 70% of the calcination process occurs here. The rest of the process takes place in the kiln.

The material then enters the kiln at the upper end. The kiln is 14 feet in diameter by 245 feet long. The material tumbles and rotates through progressively hotter zones towards the lower end or combustion source. Throughout this area, the remaining $CO_2$ is removed and the extreme heat generates other chemical reactions in the cement formation process. The low end of the kiln is where intense heat from a flame generated from either natural gas or coal combustion causes the materials to become partially molten. Temperatures of the materials reach 2700°F and the materials emerge from the kiln as glowing, red-hot "clinkers". As the material leaves the kiln and enters the clinker cooler, it is approximately 2200°F. The clinker cooler is a horizontal grate cooler through which fans force cool air over the clinker through perforated steel grates. The hot air is then recycled for heat. The clinker is then moved by drag chains, bucket elevator and conveyor belt either to the A-frame for storage or to outside storage piles.

Two baghouses control particulate emissions from the kiln process and are served by a common exhaust stack. Two gas/dust streams exit the kiln/preheater tower, one travels to the kiln baghouse and the other travels to the alkali baghouse. The kiln stream exits the preheater tower and is cooled from 1500°F to 500°F by water atomization. At this point, heavier dust particles are removed and returned to the kiln system. The stream enters the kiln baghouse where the dust is removed and the gases pass out the stack where they are monitored by the CEM and opacity monitor. The waste kiln dust consists of partially calcined kiln feed with elevated levels of alkalis and sulfur. The alkali bypass removes dust laden with volatile compounds of alkali and sulfur Otherwise, the volatile constituents would condense in the preheater tower restricting process and gas flows. There are also two baghouses that control particulate·emissions from the clinker cooler.

The cooler baghouses scrub the gas stream of particulates. The dust is returned to the cooler as product. The stack emissions are monitored by an opacity monitor. Opacity is recorded every minute for the cooler and the kiln, and monthly printouts of emissions are maintained. Opacity readings for 6-23-99 were reviewed at the time of inspection. Visible emissions of 6-7% were noted at the time of inspection (10:30 am).

**The following CEM readings were recorded at the time of inspection (10:30 am):**

| | |
|---|---|
| Kiln Stack Flow: | 108.257 dscf/m |
| CO Emissions: | 60 ppm |
| $NO_x$ Emissions: | 579 ppm |
| $SO_2$ Emissions: | 0 ppm |
| THC: | 0 ppm |
| Kiln Stack Opacity: | 7.6 % |
| Cooler Stack Opacity: | 0.51 % |

**Compliance with Permit #12BO444-2 Conditions:**

1. Annual operating hours shall not exceed 8,064 hours. Actual operating hours were 8,040.5 in 1998.

2. Emissions of $NO_x$, $SO_2$, VOC, and CO in the kiln stack are monitored on a continuous basis using a certified CEM. The CEM is certified by annual certification and quarterly audits. Records (monthly summaries) of these certifications are maintained by SWP and both of these certification procedures are conducted by different contractors. RATA information was reviewed at the time of the inspection.

3. Emissions:

**Table 3.0**

| Pollutant | PPM | Permit Lbs/hr | Permit TPY | Actual TPY |
|---|---|---|---|---|
| $NO_x$ | 550 | 656.9 | 2649 | 1708.2 |
| $SO_2$ | 200 | 332.3 | 1340 | 12.2 |
| VOC | 30 | 34.3 | 138 | 2.3 |
| CO | 135 | 98.1 | 396 | 128.7 |
| $PM_{10}$ - Kiln | n/a | 23.9 | 133 (96.4 Permit)** | 124.0 |
| $PM_{10}$ - Cooler | n/a | n/a | 44.4** | 41.3 |
| TSP | n/a | 23.9 | 96.4 | In compliance** |
| Pb | n/a | 10.8 | 43.6 | 0.450* |

Note: As per a memo *(attached)* from TomTstinic from APCD, the permit engineer, and as it states in permit condition #3, emissions estimates are based on worst case CEM data with the exception of Lead and TSP. Lead was calculated based on emission factors in AP-42 (8.6-1 which has now been changed to 11.6-9). TSP/$PM_{10}$ emissions from the kiln are based on the NSPS allowable of 0.3 lbs/ton of feed. Annual limits were then derived using 887,040 tons or feed and 8,064 hours per year. TSP/$PM_{10}$ emissions from the clinker cooler are based on the NSPS allowable of 0.1 lbs/ton of feed. On the same memo Bob Jorgenson, of APCD, Compliance and Enforcement says that "We should be using the monitoring system on the kiln stack to judge compliance. SWP Cement should be doing a 12-month rolling total of the emissions that can be checked against the yearly emission rates". SWP has provided a 12-month rolling total of emissions of CO, $NO_x$, VOC (THC), and $SO_2$ *(see attached Kiln Rolling Monthly Emissions).*

* SWP has also provided Lead emissions *(please see attached "Kiln" spreadsheet for calculated Lead emissions)*. Based on new EF's in AP-42 ($75 \times 10^{-5}$ lbs/ton of clinker) and research done by Southdown at their Knoxville facility that showed that 99.958% of the total Lead entering the kiln is retained in the kiln, the emission factor for Lead would be 0.00013 lbs/ton of dry feed. To be conservative, Southdown has used 0.001 lbs/ton of feed for an EF for Lead.

**According to Amarjit Gill, the continuous opacity monitors are used to show compliance with the NSPS limits for TSP/$PM_{10}$. ($PM_{10}$ is assumed to be 100% of TSP.) The $PM_{10}$ limits, 133.0 tons/yr for the kiln and 44.4 tons/yr for the cooler, were calculated based on the following calculations: (0.3 lbs/ton of feed) (max. feed of 887,040 tons/yr)/2000 lbs/ton = 133.0 tons/yr
(0.1 lbs/ton of feed) (max. feed of 887,040 tons/yr)/2000 lbs/ton = 44.4 tons/yr
*(See also Condition #11, below.)*

**Table 4.0 Permit #12BO444-2 Limits**

| Description | Units | Hr. Average | Hr. Max | Permit Annual | Actual Annual |
|---|---|---|---|---|---|
| **Kiln Feed (dry)** | Tons | 110 | 120 | 887,040 | 826,631 |
| **Heat Input** | MMBTU | 325 | 340 | $2.62 \times 10^6$ | $0.23 \times 10^6$ |
| **100% Coal** | Tons | 14.13 | 14.84 | 113,945 | 97,911 |
| **100% Nat. Gas** | SCF | 302,325 | 317,440 | $2{,}438 \times 10^6$ | $245 \times 10^6$ |
| **Tires** | Tons | 2.28 | 2.50 | 18,400 | 0 |

Note: As per Permit Condition #3, the actual hourly averages of the kiln feed and fuel consumption rates *(above)* are not intended to establish limits of the kiln, but are the basis for the stated emission limits.

4. Compliance with annual limits for $NO_x$, $SO_2$, VOC and CO have been determined on a rolling 12-month total by SWP Cement and they maintain a compliance record on site. *(Please see attached spreadsheets "Rolling Monthly Emissions".)*

5. A revised APEN will be filed if a significant change in emissions occurs.

6. SWP has notified the Division and BCHD of any upset conditions within 2 hours of the next working day with an explanation as to the cause and of the actions made to correct and prevent the problem.

7. The facility and control equipment are maintained and operated in a manner consistent with good air pollution control practices to minimize emissions. This was determined based on monitoring data and inspection of the facility.

8. A source compliance test was conducted to measure particulate matter in the main (kiln) stack on August 3, 1994 and on August 17, 1994 for the clinker cooler.

9. Daily kiln feed rates, fuel feed rates, clinker production, opacity measurements and operating hours are maintained. Monthly summaries are compiled and kept at the facility for at least two years. This file includes raw material consumption, quantities of coal, natural gas, and the amount of clinker produced. Clinker production numbers and emission factors from AP-42 Table 8.6-1 (now 11.6-9) are used indirectly to determine compliance with emission limits for $PM_{10}$ and Lead.

*Note: Since the permit was issued, AP-42 has been updated (1/95) and these emission factors have changed. $PM_{10}$ now states "ND" or "No Data", TSP or PM has changed from 256 lbs/ton to 250 lbs/ton and Lead has changed from 0.12 lbs/ton to $7.5 \times 10^{-5}$ lbs/ton.*

10. The permit number is marked on the kiln.

11. Source is in compliance with Regulation No. 6, Part 60, Subpart F, Standards of Performance for Portland Cement Plants including: 20% opacity from the kiln, 10% opacity from the clinker cooler, Particulate Matter does not exceed 0.3 lb/ton of dry feed from kiln and Particulate Matter does not exceed 0.1 lbs/ton of dry feed from clinker cooler. This is demonstrated by the CEM data which shows them to be under 20% from the kiln stack. Source is also in compliance with the requirements of Regulation No. 6, Subpart A, General Provisions: 60.12, 60.7, 60.63, 60.65, 60.8, 60.11 and 60.13.

12. Source is in compliance with Regulation No. 1, VI.3.f which limits $SO_2$ emission to 7 lbs/ton of material (including fuel) processed. Actual $SO_2$ emissions are calculated to be .05 lbs/ton. *(Please see attached "Kiln" spreadsheet for this calculation.)*

**Compliance Status: [ X ] In compliance [ ] Out of Compliance**

**Point 09, Clinker Storage Silos (This point is now covered under Permit #98BO0259)**

There are two clinker storage silos and one gypsum storage silo. When they make masonry cement, they use one of the clinker silos to store limestone. There were no visible emissions observed from this point during the inspection. There is a baghouse to control fugitive emissions.

**Point 10, Clinker Conveying and A-Frame Storage (This point is now covered under Permit #98BO0259)**
Drag chains transfer the cooled clinker to a conveyor belt which takes it to the A-Frame for storage. The A-frame is a huge building (200 feet wide x 388 feet long and 80 feet high at the apex) which provides covered storage for the clinker prior to being transferred to the Finish Mill. The A-frame has several hoppers underneath the clinker storage piles that funnel the clinker onto another conveyor that transfers the material to the storage silos that feed the finish mill. There was a noticeable amount of dust on top of the A-frame that should be cleaned off as a house-keeping measure. There are baghouses at the conveyor inlet, conveyor discharge and at conveyors 629-5 and 629-6 (underneath the A-frame).

**Points 11, 12, 13, Finish Mill, Cement Load-Out, and Cement Storage (These point are now covered under Permit #98BO0259)**
The finish mill is also a ball-mill. It uses steel balls to grind the clinker and small amounts of limestone into the fine, gray powder which is Portland cement. Limestone is also ground with clinker and gypsum to make masonry cement. The finish mill grinds the materials to a fineness of which 95% passes through a standard 325 sieve. The material (cement) is cooled and then pumped pneumatically into storage silos. The cement storage silos have a capacity of holding 35,600 tons of cement. SWP makes seven different types of cement. Most of the cement is shipped by bulk transport trucks which are gravity loaded from the silos. A small amount of cement (1-2%) is bagged for local customers. There were no visible emissions from either of these points. The particulate emissions for all three points are controlled by baghouses. A high efficiency separator and a product collection baghouse in the finish mill has replaced the 2 existing separators. The high effeciency separator helps achieve a better quality of cement and at the same time avoids over grinding, producing less waste (CKD and beneficiating dust have been reduced drastically).

**Point 14, Coal Unloading & Storage (Exempt)**
Coal is unloaded from railroad cars within the coal unloading building and conveyed to the coal mill. The coal mill grinds the coal but this process is totally self contained. The coal is then fed to a preheater and then to the kiln. This point used to be permit #P-10-718, but was granted an exemption from permit requirements in 1993. It should be noted that even though the point is exempt, if the thruput exceeds 100,000 tons per year of coal, a Revised APEN will need to be filed with APCD. There was no coal being unloaded during the inspection. The coal loadout for 1998 was 97,911 tons.

**Point 15, Outside Clinker Storage/Piles (Permit #84B0369F) --Cancelled**
Once Permit No. 98BO0259 is finaled, Permit No. 84BO369F will be cancelled. Clinker storage piles are now covered under the new permit. The Fugitive Dust Control Plan is also covered under the new permit.

**Points 16-24 (old points), Raw Material Acquisition- (Mining &Material Acquisition at the Lyons Quarry has Ceased).**
All mining and materials acquisition at the Lyons Quarry has ceased. The area has been revegetated and native grass is beginning to cover the hillside. Southdown now relies solely on Dowe Flats quarry for raw material. C-Pit continues to be used as the CKD disposal pit. The remainder of the pits at the Lyons site have been reclaimed. CKD storage, handling, processing, haulage and disposal and Raw Material Waste Dust or "Beneficiation Dust" is now covered under Permit #98BO0315. This new permit has not been self-certified to date because the new Twin-Screw Pugmill associated with this process has not been installed.

**Points 24A & 24 B *(Initial Permit # 98BO0292-)*- Discharge from Stacker Conveyor Belt & Underground Reclamation; and Handling of material by front-end loaders of material not handled by underground reclamation system.**
This source has not been constructed yet. A requests was made by Southdown to Headquarters on April 27, 1999 for funding for the underground reclamation system. Approval for this system is pending, but is expected soon. Material not handled by the underground reclamation system will be handled by front-end loaders.

**Point 025: Dowe Flats Quarry Operation *(Modified Initial Approval Permit # 93BO1414-F)***
The Dowe Flats quarry in now complete and in full operation. Operation of Dowe Flats began in August /September of 1997. Mining is no longer done at the Lyons plant. Currently, all the raw material comes from Dowe Flats and is conveyed over the highway to the plant via a large, enclosed conveyor system. One staff person walks the length of the conveyor belt 2-3 times per week checking the belts on the conveyor and looking for and recording problems. Blasting takes place on weekdays from 10:00 am until 2:00 pm and no blasting is allowed on the weekends. A water truck was watering the quarry area at the time of inspection. The main road into the quarry had magnesium chloride applied to it. The site consists of a total of 1600 acres, 350-380 of which is minable. No more than 90 acres will be mined and disturbed at any one time. Their current process is to mine 300 feet and then reclaim 300 feet as they go along. The primary crusher and dust collector is equipped with a wind anemometer that shuts down the crusher when winds reach 30 miles per hour for longer than 15 minutes.

Muck piles (raw material being mined) are currently watered by a water wagon, but Southdown is in the process of installing a sprinkler system similar to the one installed at the Lyons plant for the C-Pit or waste dust disposal pit. A sprinkler system will maintain moisture in the material from the time of initial loading until it has been conveyed and dropped from the stacker into the Lyons plant. At the time of inspection, pipe lines were being laid for the sprinkler system. There was some dust from dumping the materials in the pit at the time of inspection. However, visible emissions lasted for a period of 30 seconds and were not off-property (didn't leave the pit they were working in). A berm has been built along the quarry all the way up County Road 53 and will be 40' high at the southwest end near the crushing plant. Southdown is now using only high calcium limestone in order to decrease CKD. Shale is used in cement to spread out the limestone and make the cement go further.

**Compliance with Initial Permit 93BO1414F:**

1. Initial approval of this permit has been issued. Southdown self-certified this permit in January. When finaled, all previous versions of this permit shall be canceled.

2. If more than one provision, the most stringent provision shall apply.

3. This permit modification is related to modifications of permits 94BO593 (primary crusher), 98BO0259 (finish mill), 99BO0292 (raw materials and reclamation), and 98BO0315 (CKD handling). No other equipment at this facility is affected by this project. Net changes in emissions are below major modification thresholds.

4. Particulate Emission Control Plan for Mining Activities per Appendix A:
   a. **Mining Activities** - Visible emission do not exceed 20% (for more than 6-minutes) and there was no off-property transport of visible emission at the time of inspection. Materials are watered in the quarry prior to loading and dumping. Southdown continues to work on the installation of a sprinkler system in the Dowe Flats quarry.
   b. **Haul Roads** - There was no off-property transport of visible emissions from on-site haul roads. There were some fugitive emissions from the roads into the quarry, but it was not off property and a water truck came over and watered. Magnesium Chloride has been applied to the main haul roads in the quarry.
   c. **Haul Trucks** - There was no off-property emissions from the trucks.

   *Control Measures*

   1.) Adequate soil moisture must be maintained in topsoil and overburden to control emissions during removal. A water truck was watering the pit at the time of inspection. Southdown is in the process of installing a sprinkler system to adequately maintain moisture in the pit and control fugitive emissions. At the time of inspection, material was very wet. Moisture content is about 8-10% on average.
   2.) Topsoil shall be revegetated within one year. Quarry operations are still in the first dig or first 90 acres. As soon as the first dig is completely mined, they will begin reclaiming the 90 acre pit. Only 90 acres can be open at one time.
   3.) Emissions from material handling shall be controlled by watering with a sprinkler system. Southdown is in the process of installing the sprinkler system.
   4.) Fugitive dust activities are ceased when wind speeds reach 30 miles per hour. The control room gets a constant reading of wind speeds from its anemometer. The process shuts down when wind speeds reach 30 mph.
   5.) Speed limit signs are posted and limit speeds to no more than 35 miles per hour.
   6.) Haul roads are treated with magnesium chloride as needed. Invoices are kept as documentation of applications.
   7.) Drills are equipped with bag collectors to control emissions.
   8.) Sequential blasting is employed.
   9.) Sequential extraction of material is done to keep the disturbed areas to a minimum. Southdown works on 90 acres at one time. The current plan calls for mining 300 feet at one time and at the same time reclaiming 300 feet.
   10.)A reclamation plan has been filed with the Colorado Division of Minerals and Geology and Chris Tobey of the Boulder County Land Use Commissions receives copies of the annual reclamation report.

5. Major stationary source requirements will apply at any time net emission changes equal or excess the major modification significance solely by relaxation of any permit condition.

6. Visible emissions did not exceed 20% during inspection. Emissions were noted from the dumping of the materials into the trucks and some emissions off of the haul roads in the current mining pit. However, these emissions were below 20% and were not off property.

7. Southdown submitted an operating and maintenance plan with a proposed record keeping format with the Self-Certification. The plan follows the suggested outline of the Division and includes control measures such as installing the new sprinkler system and maintaining moisture content of the soil. They have not heard from the Division yet if it is approved.

8. Process and Emission Rates:

   Process Limits

   | Permit Limit | Actual Process |
   | --- | --- |

| | |
|---|---|
| 3,200,000 tons/year | 2,562,632 tons/year |
| 400,000 tons/month | 213,552 tons/month* |
| 25,000 tons/day | 12,320 tons/day** |

*Based on 12 months of operation.
**Based on 4 days per week of operation.

Emission Limits – Dowe Flats Quarry Operations

| | Permit Limit | Actual Emissions* |
|---|---|---|
| PM | 134.2 tons/year | 42.57 tons/year |
| $PM_{10}$ | 58.4 tons/year | 17.00 tons/year |
| | 7.3 tons/month | 1.42 tons/month |
| | 916 lbs/day | 163 lbs/day |

*Actual emissions based upon 4 days per week, 12 month per year of operation.
** Emissions are based on both Quarry & Crusher Operations. See "Dowe Flats Quarry & Crusher Tons" for emission calculations.

Emission Limits – Disturbed Area at Lyons Quarry

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 19.0 tons/year | 11.40 tons/year* |
| $PM_{10}$ | 9.4 tons/year | 5.64 tons/year * |

*Emissions based on .380 ton/ac-yr for TSP and 0.187 ton/ac-yr for PM-10.
See "Lyons Quarry Emissions" attached.

9. Compliance with the above emission limits is demonstrated by being under their process limits and following the control measures listed on Appendix A.

10. Non-applicable – A revised APEN is not required at this time.

11. There is no mining at the Lyons Quarry. Only reclamation activities at this point. Currently, there are about 50 acres that are unvegetated. CKD disposal (permitted separately) occurs at the Lyons Quarry in the C-Pit.

12. Southdown submitted the self-certification form January 1999.

13. This source is subject to Regulation No. 3, Part C, Operating Permits.

14. Changes to this permit classify it as a synthetic minor modification of a major stationary source.

15. Upset conditions are reported as required in the Common Provisions Regulation Part II, Subpart E. Written notice is submitted to the Division.

**Compliance Status: [X] In compliance [ ] Out of Compliance**

**Points 026 & 027, Modified Initial Approval Permit No. 94BO593**

Subject Equipment:
**Point 026** Belt conveyor system, rated at 800 tons per hour, approximately 2.5 miles long, with 8 transfer points. The entire system is enclosed in a corridor and the transfer points are fully enclosed to minimize emissions of particulate matter. This belt conveyor system is used to transport limestone from the Dowe Flats quarry to the Lyons cement plant.

**Point 027** One (1), Make, Model, and S/N: not available, primary crusher, rated at 800 tons per hour for crushing of limestone mined in the Dowe Flats quarry. This is equipped with a pneumatic exhaust system vented through a filter baghouse to control emissions of particulate matter.

**Compliance with Initial Permit 94BO593:**

1. Initial approval has been given for this permit. Southdown self-certified this permit in January of this year. When finaled all previous versions of this permit shall be canceled.

2. If more than one provision, the most stringent provision shall apply.

3. This permit modification is related to modifications of permits 93BO1414 (Dowe Flats Quarry), 98BO0259 (finish mill), 98BO0292 (raw materials and reclamation), and 98BO0315 (CKD handling). No other equipment at his facility is affected by this project. Net changes in emission are below major modification thresholds.

4. Visible emissions did not exceed 20% during inspection. Crusher was not operating at the time of initial inspection. A follow-up visit showed no emission in excess of 20% nor were they off property. Emissions from the truck loadout into the crusher was contained in the crusher housing.

5. Major stationary source requirements will apply at any time net emission changes equal or exceeds the major modification significance solely by relaxation of any permit condition.

6. Crusher and conveyor system process and emission rates:

Process Limits

| | Permit Limit | Actual Process |
|---|---|---|
| | 1,050,000 tons/year | 860,367 tons/year |
| | 400,000 tons/month | 71,697 tons/month |

Emission Limits:

| | Permit Limits | Actual Emissions* |
|---|---|---|
| PM | 0.14 tons/year | 0.11 tons/year |
| $PM_{10}$ | 0.07 tons/year | 0.055 tons/year |
| | 0.01 tons/month | 0.005 tons/month |

*See "Dowe Flats Quarry and Crusher Tonnage" for emission data.
Emissions are based on tons processed by crusher, emission rate and control efficiency of dust collector.

7. The conveyor system is subject to NSPS Regulation No.. 6, Part A, Subpart F, Standard of Performance for Portland Cement Plants. Applicable specific emission standards:
   a. Gases discharged into the atmosphere shall not exhibit 10% opacity. A performance test done in August 1998 showed the highest average opacity to be 1.04%. *See attached copy of opacity readings.*

   The primary crusher is subject to NSPS Regulation No. 6, Part A, Subpart OOO, Stands of Performance for Non- Metallic Mineral Processing Plants. Applicable specific emission standards:
   b. Gases discharged into the atmosphere shall not exhibit 7% opacity or greater. A performance test showed 0.23% opacity.
   c. Gases shall not contain particulate matter in excess of 0.05 grams per dry standard cubic foot. Results showed an average of 0.00 gr/dscf. (Dust Collector #1 controls particulates from the crusher)- See Attached.
   d. Fugitive emissions shall not be greater than 10%. Opacity results from August 1998 show 0% opacity from the crusher (Dust Collector #1). At the time of the follow-up inspection, opacity was also 0%.

   Additional requirements:
   e. At all times, equipment shall be maintained and operated in a manner consistent with good air pollution control practices for minimizing emissions. Southdown walks the conveyor belt system 2-3 times per week and notes any problems they find. Problems are corrected as soon as possible and records are maintained with details of problems noted and when and what was corrected.
   f. No article, machine or equipment is used to conceal emissions that would otherwise violate a standard.
   g. Written notification of initial startup dates has been submitted to the Division.
   h. Records of startup, shutdown and malfunctions are maintained.
   I. Notification of opacity observations are submitted to the Division as required.
   j. Excess Emission and Monitoring Reports are submitted to the Division as required.
   k. Performance tests are conducted as required. The last test was performed December 1997.
   l. Compliance with opacity standards is determined per 60.11 (Method 9).

8. A revised APEN is not required at this time.

9. The emission sources are equipped with emission control device capable of controlling the emissions. Transfer points on the conveyor belt are equipped with baghouses and the crusher is enclosed in a building and also equipped with a baghouse. The operating parameters of this equipment has been identified and submitted to the Division for approval.

10. Southdown submitted the self-certification for this permit in January 1999.

11. Available manufacturer, model number and serial numbers of the subject equipment has been provided to the Division.

12. The AIRS ID numbers are clearly marked on the equipment.

13. This source is subject to Regulation No. 3, Part C, Operating Permits.

14. Changes to this permit classify it as a synthetic minor modification of a major stationary source.

15. Compliance tests have been conducted to measure emission rates to demonstrate compliance with emission limits. Southdown submitted a compliance test manual prior to testing and received approval from the Division prior to conducting any testing. Tests were conducted in December 1997.

16. Southdown submitted an operating and maintenance plan and a proposed record keeping format prior to the compliance tests in December 1997.

17. All upset conditions and breakdowns are reported to the Division and Boulder County Health Department. Written follow-ups with notes on corrections are submitted to the Division within 2-hours of the start of next business day.

18. Manufacturing is subject to New Source Performance Standards of Regulation No. 6, Part B, Subpart III. Southdown calculates emissions based on NSPS. See previous emission calculations.

**Compliance Status: [ X ] In compliance [ ] Out of Compliance**

<u>**Point 31,39: High Efficiency Separator & Finish Mill System *(Initial Approval Permit #98BO0259)*-( Replaces Permit #84BO369F)**</u>
This is a system wide permit covering the Finish Mill System.

**Compliance with Permit Conditions:**

1. Initial approval has been given for this permit. Start-up for this source was March 23, 1999, so Southdown has until September 23, 1999 to self-certify this permit . When finaled all previous versions of this permit shall be cancelled. This includes permit 84BO369F (outdoor clinker storage).

2. If more than one provision, the most stringent provision shall apply.

3. This permit modification is related to modifications of permits 93BO1414 (Dowe Flats Quarry), 98BO0593 (primary crusher and conveyor system), 98BO0292 (raw materials and reclamation), and 98BO0315 (CKD handling). No other equipment at his facility is affected by this project. Net changes in emission are below major modification thresholds.

4. Source is constructed. The finish mill system, also know as the high efficiency separator, began operation March 1999.

5. Major stationary source requirements will apply at any time net emission changes equal or exceeds the major modification significance solely by relaxation of any permit condition.

6. Visible emissions did not exceed 20%. At the time of inspection, emissions were 0%. There was dust noted inside the building that houses the finish mill, however it was not outside.

7. Since this is the first 12-months of operation, compliance is demonstrated with the both the monthly and yearly process and emission rate limits. After 12-months, compliance will be done on a rolling 12-month total. The following points are covered by this permit and include April through July process rates:

**009 - Clinker and Gypsum/Storage Silos**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Clinker Handled | 600,000 tons/year | 155,997 tons/year |
| | 75,000 tons/month | 38,999 tons/month |
| | 4,000 tons/day | 1,279 tons/day |

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 7.02 tons/year | see below ** |
| $PM_{10}$ | 3.51 tons/year | |

0.33 tons/month
22.0 lbs/day

**Emissions not calculated. It is assumed to be under permit limits since process amounts are well under process limits.

**010 - Sheltered Clinker Storage w/Underground Reclamation**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Clinker Handled | 600,000 tons/year | 108,486 tons/year |
| | 75,000 tons/month | 27,121 tons/month |
| | 5,500 tons/day | 889 tons/day |

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 0.82 tons/year | see below ** |
| $PM_{10}$ | 0.41 tons/year | |
| | 0.05 tons/month | |
| | 7.5 lbs/day | |

**Emissions not calculated. It is assumed to be under permit limits since process amounts are well under process limits.

**015 - Outdoor Clinker Storage and Front-End Loader Handling**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Clinker Stored | 15,000 tons/year** | 12,805 tons/year* |
| Clinker Handled | 22,500 tons/year** | 26,143 tons/year |
| | 5,500 tons/day | 214 tons/day |

*Maximum amount of clinker stored on site in 1998.

** These limitations are incorrect. The latest draft of the Title V Permit allows 120,000 tons/year for clinker stored, and 180,000 tons/year for clinker handled. The 22,500 ton limit would only allow Southdown to move material 4 days a year-rather than the 7 days a week requested for normal operation. "Title V Permit Limits Attached."

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 3.35 tons/year | see below ** |
| $PM_{10}$ | 1.30 tons/year | |
| | 0.16 tons/month | |
| | 78.0 lbs/day | |

**011 - Cement Finish Mill and Bucket Elevator**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Fresh Feed Into Mill | 600,000 tons/year | 155,997 tons/year (518,399 ton/year)* |
| | 75,000 tons/month | 38,999 tons/month (43,199 tons/month)* |
| | 3,000 tons/day | 1,279 tons/day (2,591 tons/day)* |

*The new Finish Mill Operations began March 23, 1999. Numbers in (parenthesis) are estimates for a full year of production based on the first quarter.

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 15.93 tons/year | see below ** |
| $PM_{10}$ | 7.96 tons/year | |
| | 0.71 tons/month | |
| | 48.0 lbs/day | |

**Emissions not calculated. It is assumed to be under permit limits since process amounts are well under process limits.

**031 - High Efficiency Separator**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Cement Produced | 600,000 tons/year | 155,997 tons/year (518,399 ton/year)* |
| | 75,000 tons/month | 38,999 tons/month (43,199 tons/month)* |
| | 3,000 tons/day | 1,279 tons/day (2,591 tons/day)* |

*High Efficiency Separator began operation March 23, 1999. Numbers in (parenthesis) are estimates for a full year of production based on the first quarter.

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 35.86 tons/year | see below ** |

| | | |
|---|---|---|
| $PM_{10}$ | 17.93 tons/year<br>1.61 tons/month<br>107.0 lbs/day | |

**Emissions not calculated. It is assumed to be under permit limits since process amounts are well under process limits.

**013 - Cement Storage Silos/Loadout**

*Process Limits*

| | Permit Limit | Actual Process |
|---|---|---|
| Cement Handled | 600,000 tons/year<br>75,000 tons/month<br>3,000 tons/day | 171,330 tons/year<br>42,832 tons/month<br>1,586 tons/day |

*Emissions*

| | Permit Limit | Actual Emissions |
|---|---|---|
| PM | 11.39 tons/year | see below** |
| $PM_{10}$ | 5.70 tons/year<br>0.72 tons/month<br>43.0 lbs/day | |

**Emissions not calculated. It is assumed to be under permit limits since process amounts are well under process limits.

8. The equipment and activities in this process system are subject to NSPS, Regulation No. 6, Part A, Subpart F, Standards of Performance for Portland Cement Plants. The following standards are required:
   a. At all times, the control equipment is operated in a manner consistent with good air pollution control practices determined by review of operating and maintenance procedures, records on file and inspection of source.
   b. No article, machine or equipment is used to conceal an emission that would otherwise be a violation.
   c. Written notification of start-up was submitted to the Division.
   d. Records of start-ups, shutdowns and malfunctions are maintained.
   e. Written notification of opacity observations will be submitted to the Division quarterly once the permit is finaled.
   f. Excess Emission and Monitoring System Performance is submitted to the Division quarterly.
   g. Performance tests will be conducted in early September. Testing protocol has been submitted to the Division and approved by Tom Lovell of CDPH&E.
   h. Compliance with opacity is demonstrated according to 60.11(Method 9)

9. Particulate emission control measures:
   1. Roads and edges of clinker piles are watered at least once per day. There was a visible crust on the surface of the piles.
   2. Traffic on and around the storage piles is limited.
   3. Height of fall material is minimized and adjusted when necessary and dust extractor is in close proximity to the emission source.
   4. Vehicle traffic on unpaved surfaces is limited to established roadways. Roadways are watered as necessary.
   5. Clinker is reclaimed as soon as possible. At this time, there is a shortage of cement so there is limited storage.
   6. Paved areas are kept clean with three vacuum trucks that operate about eight hours a day.
   7. Activities are suspended when wind speeds reach or exceed 30 mph.
   8. Spillages are cleaned up as soon as they happen.

At the request of BCHD and APCD, Southdown revised their fugitive dust control plan to address the waste dust disposal area and fugitive emissions throughout the plant.

***Revised Dust Control Plan (Submitted by Southdown May 21, 1998) - (See attached copy)***

<u>New Dust Pit</u>:

The new dust storage pit (C-Pit) is closer to the plant and has reduced the travel distance by 75%. The C-Pit is approximately 20 acres and is 20 to 100 feet below the existing ground level. Of the 20 acres, only 7 acres at a time will be used for dust storage. The remaining 13 acres are to be covered with shale. A sprinkler system was installed and operating at the time of inspection. The sprinklers are set on an automatic timer and keep the dust disposal area wet.

<u>Fugitive Controls/Monitoring/Record Keeping</u>:

Dust from the baghouse is controlled by a pelletizing process (pugging) where by the dust is exposed to waters sprays as the dust exists the silos and passes along a paddle screw conveyor forming pebble size pellets. The pellets have a moisture content of 20-25% which are dropped into a haul truck for direct transport to the dust pit. The company has increased the size of the haul trucks from 50 ton capacity to 95 ton capacity to reduce the number of trips taken. Trucks are only loaded 50-75% capacity.

Magnesium Chloride will be applied to heavily traveled haul roads including the road into the dust pit throughout the summer, as needed.

At the time of inspection, the Magnesium Chloride was clearly present and controlling dust from the main haul roads. Additionally, untreated roads and travel areas throughout the plant will be watered daily. A water truck was wetting the roads and areas around the plant at the time of inspection.

There will usually be someone at the plant certified to take visible emission readings. Currently two employees are certified to take opacity readings including Steve Mossberg. A Daily Opacity Observation Sheet is filled out and stored in binder with a photo of the plant.

The dust monitoring program involves a day shift, production shift supervisor, making daily rounds of the plant inspecting for dusty conditions. When an area is identified that needs attention, every effort is made to correct the situation. If the scope of the problem is larger, a work request will be written through a computerized maintenance system to rectify the problem. This information is logged daily and kept by the compliance manager. There is plant personnel on site 24-hours a day to provide regular maintenance on all dust collection equipment. Plant watering records are maintained, indicating the frequency and number of gallons applied. Daily photos of the facility are also taken with a date-back camera to document compliance. Watering and maintenance records were reviewed at the time of inspection and the photos were observed.

***Revised Dust Control Plan April 14, 1999***
On March 29, 1999 at the request of Dave Ouimette of CDPH&E, Southdown revised their dust control plan to include the following: seven day a week watering of the plant and newly reclaimed quarry; watering the quarry reclamation area until grass seed begins to grow; adding extra dust suppressant to the loads of water used to water the plant; applying Magnesium Chloride to all unpaved haul roads; and adding two to three additional applications of Magnesium Chloride throughout the summer and fall according to conditions on the road.

At the time of inspection the haul roads were wet and treated with dust suppressant and the quarry reclamation area was beginning to sprout native grass.

10. Non-applicable - A revised APEN is not required at this time.

11. The emission sources are equipped with emission control devices as specified in Attachment A in permit. The six points covered in this facility wide permit each have a baghouse for controlling emissions. Southdown has records of inspections and monitoring that has been done. They are still working on identifying the operating parameters that will be incorporated into the final permit.

12. Permit has been self-certified. Self-certification was submitted in January 1999.

13. The manufacturer, model number and serial number of the subject equipment has been provided to the Division. The new high efficiency separator is a Humboldt-Wedag, model# SKS-210.

14. The AIRS ID numbers are marked on the subject equipment.

15. This source is subject to the provision of Regulation No. 3, Part C, Operating Permits (Title V).

16. The above changes to the facility is classified as a synthetic minor modification of a major stationary source.

17. Compliance tests shall be conducted to measure emission rates in order to demonstrate compliance. A compliance test manual was submitted to the Division and approved by Tom Lovell. The tests are being scheduled for early September.

18. An operating and maintenance plan for all control equipment and proposed record keeping format will be submitted along with the Self-Certification. The plan specifies a maintenance schedule, reducing traffic around the clinker piles, keeping the clicker wet down, watering roads to clinker piles, keeping paved areas swept ect..

19. This source is subject to Upset Conditions and Breakdowns. All upset conditions are reported to APCD and to BCHD by the start of the next working day via FAX. Written notice is sent to the division.

20. Manufacturing at the is facility is subject to NSPS Regulation No. 6, Part B, Subpart III - Standards of Performance for New Manufacturing Processes. This part of the regulation governs particulate emissions. The scheduled stack test will demonstrate compliance with this condition.

**Points 40A,40B,41,42,49: Storage, Handling, Processing, Haulage and Disposal of Cement Kiln Dust and Raw Materials Waste Dust "Beneficiation Dust". *(Initial Approval Permit #98BO0315)* -Not Operating**

This permit has not been self-certified to date because the Twin-Screw Pugmill associated with this permit is not operating. The Twin-Screw Pugmill which will provide better pugging of the CKD and Beneficiation dust is under construction.

**Source Compliance Status: [ X ] In compliance [ ] Out of Compliance**